先进计算与数字工程研究中心

PATTERN RECOGNITION AND MACHINE LEARNING

Lecture 10 – Clustering: Distance, Evaluation, and Classical Algorithms



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Clustering syllabus blocks and timing

Syllabus block	Time	Driving question
Supervised versus unsupervised	0.5 h	What can be claimed without labels?
Distance and similarity	0.5 h	What does near mean?
Clustering evaluation	0.5 h	Which evidence supports a partition?
Classical algorithms	1.5 h	Which structural assumption matches the data?



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The same points admit several partitions

Observation

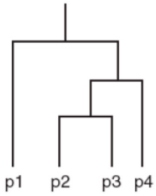
A cloud of points can be split by location, density, shape, or hierarchy.

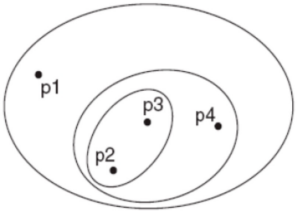
Question

Which grouping helps the downstream decision?

Claim

A partition is not ground truth merely because it looks clean.





A clustering analysis starts with the meaning of similarity

- Define one object and select the features describing it.
- Choose scaling and distance according to the task.
- Fit a method whose cluster shape matches the question.
- Inspect stability and describe clusters in the original units.

Supervised learning receives target labels

Core idea

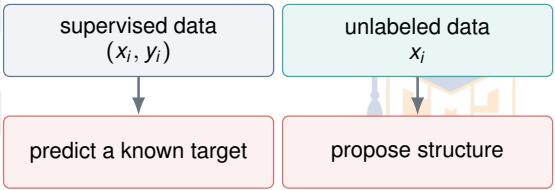
Training pairs contain both inputs and a named outcome.

Concrete case

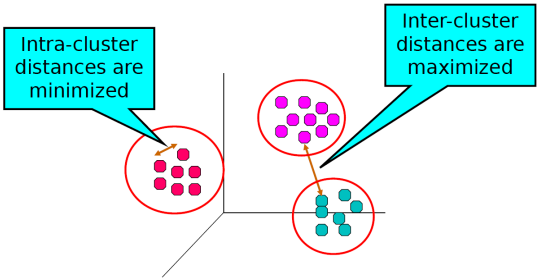
A classifier learns to predict a known defect label.

Watch for

Labels can still be noisy or biased.



Clustering groups similar objects



- Within-cluster similarity should be high and between-cluster similarity low under a chosen measure.
- Products with comparable purchase patterns may form a segment.
- The statement is incomplete until similarity is defined.

Three elements define a clustering task

Start from evidence

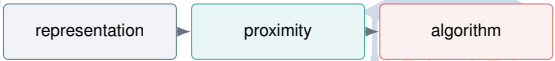
Changing units can change distances even when the algorithm is fixed.

Model statement

Representation, proximity, and algorithm must be specified together.

Failure mode

A model name alone does not reproduce a partition.



Changing any one element can change the partition.

The unit of analysis comes first

What changes

One row must represent the entity that will receive a cluster label.

What can break

Clustering line items can accidentally segment purchase occasions instead of people.

Try this

Aggregate transactions to customers before customer segmentation.

Features encode the notion of relevance

Lens	Evidence
Principle	Include variables that express the similarity needed by the decision.
Case	Service usage frequency may matter more than account identifier.
Risk	Identifiers create arbitrary separation and should usually be excluded.



A partition assigns every object once

$$C_1 \cup \cdots \cup C_K = \{1, \ldots, N\}, \qquad C_j \cap C_k = \varnothing$$

Read it

K-means returns a hard partition.

Boundary

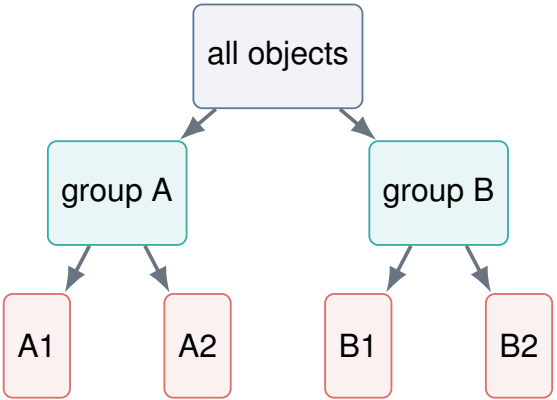
Noise-aware methods may leave some objects unassigned.

Hard and soft memberships answer different questions

- Hard clustering returns one label; soft clustering returns a membership vector.
- A customer can be 0.6 value-seeking and 0.4 convenience-seeking under a mixture model.
- Soft membership is not automatically a calibrated business probability.



Hierarchical clustering returns many resolutions



Start from evidence

Cut high for broad families or low for fine subtypes.

Model statement

A dendrogram represents nested groups rather than one fixed K.

Failure mode

The hierarchy depends on linkage and distance choices.

Center-based clusters prefer prototypes

What changes

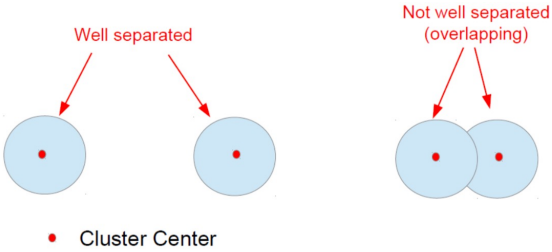
A cluster is summarized by a center and points stay near it.

What can break

Nonconvex shapes can be split incorrectly.

Try this

K-means uses arithmetic means as prototypes.



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Contiguity clusters follow connected chains

Core idea

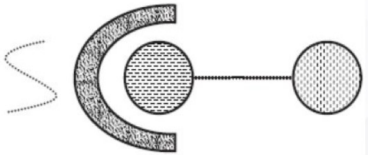
Membership can be supported by a path of nearby observations.

Concrete case

A curved river-shaped group can stay connected.

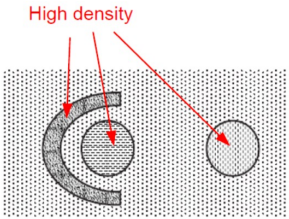
Watch for

Single-link methods may chain through accidental bridges.



(c) Contiguity-based clusters. Each point is closer to at least one point in its cluster than to any point in another cluster.

Density clusters separate high-density regions



(d) Density-based clusters. Clusters are regions of high density separated by regions of low density.

Core idea

Dense regions are divided by areas with fewer observations.

Concrete case

DBSCAN can recover irregular shapes and label sparse cases as noise.

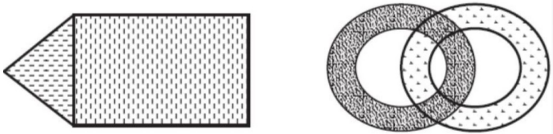
Watch for

One global density threshold struggles when densities vary strongly.

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Conceptual clusters follow shared properties

- Membership may be defined by a semantic description rather than only geometric nearness.
- Documents can cluster by topic despite different word counts.
- The representation must encode the intended concept.



(e) Conceptual clusters. Points in a cluster share some general property that derives from the entire set of points. (Points in the intersection of the circles belong to both.)

Euclidean distance measures straight-line separation

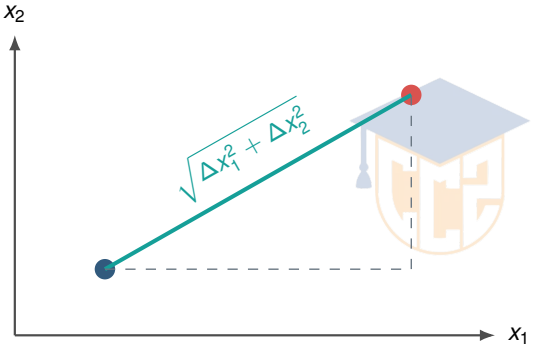
$$d_2(x, z) = \sqrt{\sum_j (x_j - z_j)^2}$$

Read it

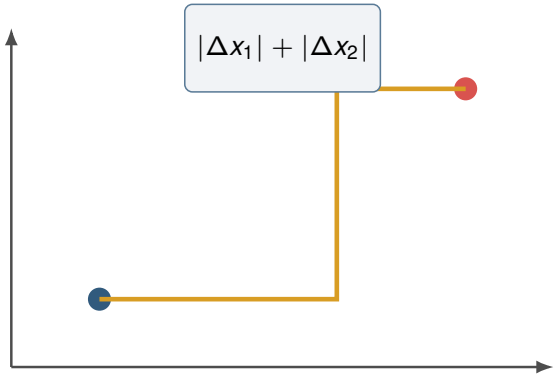
Appropriate for scaled continuous coordinates with isotropic meaning.

Boundary

A large-unit feature dominates the sum of squares.



Manhattan distance adds coordinate moves



$$d_1(x, z) = \sum_j |x_j - z_j|$$

Read it

Useful when differences accumulate independently across dimensions.

Boundary

Rotating the coordinate system changes this distance.

Minkowski distance connects common norms

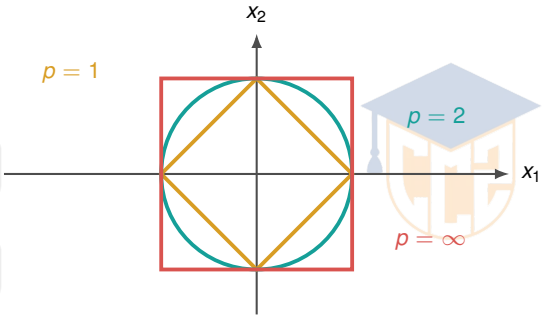
$$d_p(x, z) = \left(\sum_j |x_j - z_j|^p \right)^{1/p}$$

Read it

$p = 1$ gives Manhattan and $p = 2$ gives Euclidean.

Boundary

Large p emphasizes the largest coordinate gap.



Standardization equalizes numeric scales

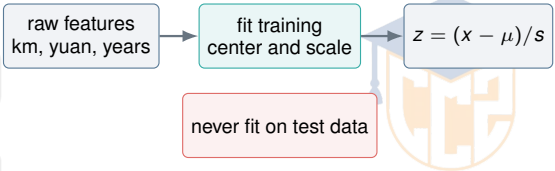
$$z_{ij} = \frac{x_{ij} - \mu_j}{s_j}$$

Read it

Age and annual spending should not compete through units alone.

Boundary

Fit μ_j and s_j on training or reference data only.



Robust scaling protects against extreme values

- Use medians and interquartile ranges when marginal outliers are strong.
- Income with a few extreme customers can distort mean and variance.
- Robust scaling does not remove multivariate anomalies.

Min-max scaling fixes a visible range

$$x' = (x - x_{\min})/(x_{\max} - x_{\min})$$

Read it

Convenient when bounded features have stable limits.

Boundary

A new extreme value can move the entire mapping.

Cosine similarity compares direction

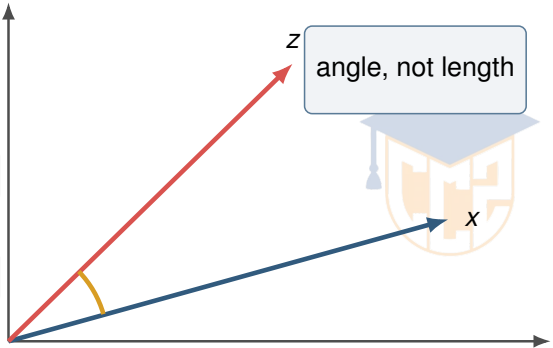
$$s_{\cos}(x, z) = \frac{x^T z}{\|x\| \|z\|}$$

Read it

Document word profiles can be similar despite different lengths.

Boundary

Zero vectors make the expression undefined.



Correlation similarity removes location and scale

Lens	Evidence
Principle	Compare centered profiles rather than raw magnitudes.
Case	Two sensors with parallel temporal patterns but different baselines can be grouped.
Risk	Short noisy profiles can show unstable correlations.

Jaccard similarity focuses on shared presences

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

Read it

Compare baskets of purchased product categories.

Boundary

Shared absences are ignored by design.

Hamming distance counts mismatches

$$d_H(x, z) = \sum_j \mathbf{1}(x_j \neq z_j)$$

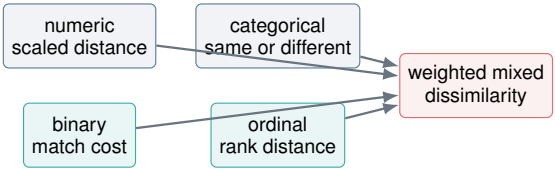
Read it

Compare fixed-length binary or categorical codes.

Boundary

Different features may need different mismatch costs.

Mixed data need mixed dissimilarity



Start from evidence

Customer profiles may include age, plan category, and yes-no preferences.

Model statement

Compute type-appropriate component distances and combine them with explicit weights.

Failure mode

Integer-encoding categories creates fake order and spacing.

Missing values change pairwise evidence

What changes

Normalize distance by the features observed for each pair.

What can break

Simple zero filling can create artificial similarity.

Try this

Two customers compared on only one field have less reliable proximity.

A distance matrix makes assumptions visible

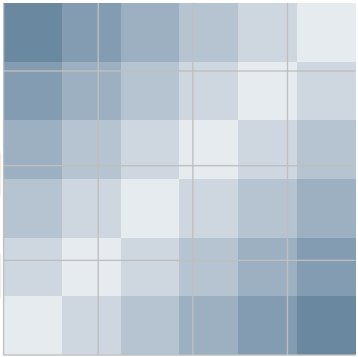
$$D_{ij} = d(x_i, x_j)$$

Read it

Inspect symmetry, diagonal values, and the empirical range.

Boundary

Storing all pairs costs quadratic memory.



zero diagonal

symmetric

pairwise view

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Distance choice is testable

- Run a controlled comparison across plausible metrics and inspect stability.
- If clusters change completely between Euclidean and cosine, report the ambiguity.
- Choosing the prettiest plot after many tries overstates evidence.

Evaluation asks three different questions

Start from evidence

A compact segmentation may still be useless for intervention.

Model statement

Compactness, agreement with reference labels, and downstream usefulness are distinct.

Failure mode

No single index answers all three questions.

External evaluation needs reference labels

What changes

Compare the found partition with labels not used during clustering.

What can break

Reference labels may encode a different concept than the clustering goal.

Try this

A known species label can audit an unsupervised iris partition.

A contingency table aligns clusters and classes

Core idea

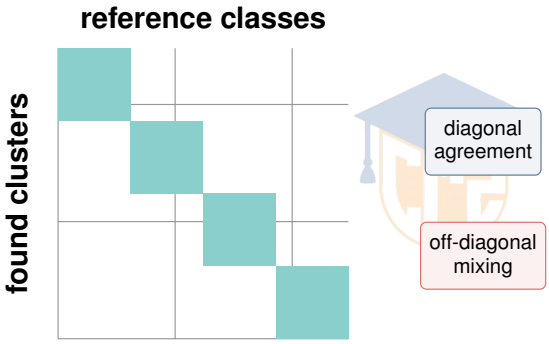
Rows represent found clusters and columns represent reference classes.

Concrete case

Concentration in a few cells indicates strong correspondence.

Watch for

Cluster numbers are arbitrary and should not be compared directly to class numbers.



Pair-count notation supports several indices

Core idea

Count pairs placed together or apart by two partitions.

Concrete case

Use a, b, c, d for the four agreement-disagreement combinations.

Watch for

Different texts swap the letters; define them before using a formula.

	same in U	different in U
same in V	a	b
different in V	c	d

choose which agreements count

Rand index counts both kinds of agreement

$$\mathrm{RI} = \frac{a + d}{a + b + c + d}$$

Read it

Pairs together in both and apart in both are rewarded.

Boundary

Many true-negative pairs can dominate when clusters are numerous.

Adjusted Rand removes chance agreement

$$\mathrm{ARI} = \frac{\mathrm{RI} - E[\mathrm{RI}]}{\max(\mathrm{RI}) - E[\mathrm{RI}]}$$

Read it

A random partition has expected score near zero under the adjustment model.

Boundary

Negative values can occur when agreement is below chance expectation.

Pairwise Jaccard ignores shared separation

$$J = \frac{a}{a+b+c}$$

Read it

Reward pairs joined by both partitions.

Boundary

Do not confuse pairwise Jaccard with feature-set Jaccard.

Fowlkes-Mallows balances pair precision and recall

$$\mathrm{FMI} = \sqrt{\frac{a}{a+b}\frac{a}{a+c}}$$

Read it

It is the geometric mean of pairwise precision and recall.

Boundary

A high value still does not explain which clusters are mixed.

Purity can reward too many clusters

$$\text{purity} = \frac{1}{N} \sum_{k} \max_{j} |C_k \cap L_j|$$

Read it

Putting each sample in its own cluster gives purity one.

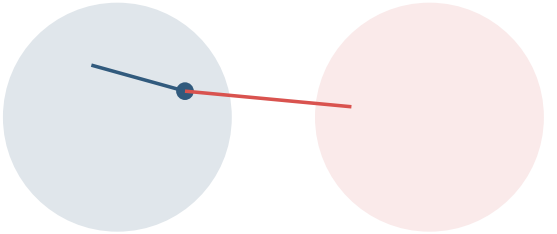
Boundary

Always report the number and size distribution of clusters.

Internal evaluation uses geometry only

- Assess compactness and separation without reference labels.
- Useful for model selection when no external labels exist.
- It inherits the chosen distance and shape preference.

Silhouette compares own and neighboring clusters



$a(i)$: own-cluster mean

$b(i)$: nearest-other mean

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

Read it

$a(i)$ is own-cluster mean distance and $b(i)$ the nearest-other mean.

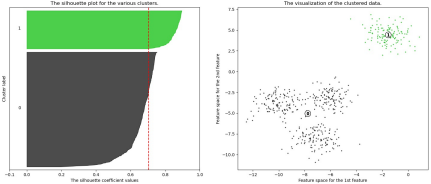
Boundary

Singleton clusters need a defined convention.

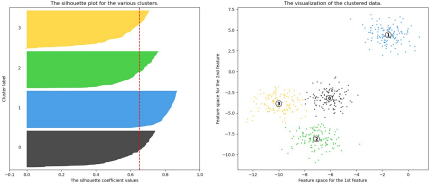
Read a silhouette profile, not only the mean

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$K = 2$



$K = 4$



What changes. Width shows sample scores; band height shows cluster size.

What can break. Sort profiles consistently and label cluster sizes.

Try this. A high mean can hide one small cluster with many negative cases.

Calinski-Harabasz rewards between-cluster dispersion

$$\mathrm{CH} = \frac{B_K/(K-1)}{W_K/(N-K)}$$

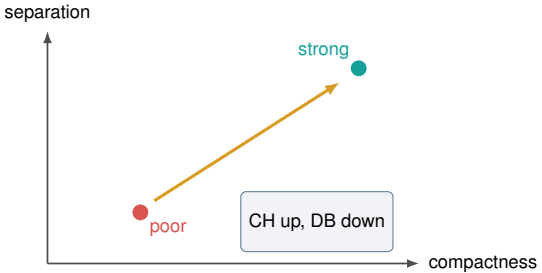
Read it

Higher values indicate separated compact groups under variance geometry.

Boundary

Scores are not comparable across unrelated datasets.

Davies-Bouldin uses worst neighboring similarity



$$\mathrm{DB} = \frac{1}{K} \sum_{k} \max_{j \neq k} \frac{S_k + S_j}{M_{kj}}$$

Read it

Lower is better when clusters are compact and centers separated.

Boundary

It favors center-like clusters and can misjudge curved shapes.

Stability tests perturb the analysis

- Resample cases, change seeds, or slightly change preprocessing and compare partitions.
- Use ARI between repeated fits after matching sample sets.
- A stable but useless partition is still not valuable.



Describe clusters using their original features

Start from evidence

A segment called high value should show a reproducible value pattern.

Model statement

Summarize original variables, representative cases, and uncertainty for each cluster.

Failure mode

Naming clusters after seeing outcomes can invite storytelling bias.

K-means minimizes within-cluster squared error

$$J = \sum_{k=1}^{K} \sum_{i:z_i=k} \|x_i - \mu_k\|_2^2$$

Read it

Each point pays squared Euclidean distance to its assigned centroid.

Boundary

The objective encodes spherical equal-scale preference.

Assignment chooses the nearest centroid

$$z_i \leftarrow \arg\min_k \|x_i - \mu_k\|_2^2$$

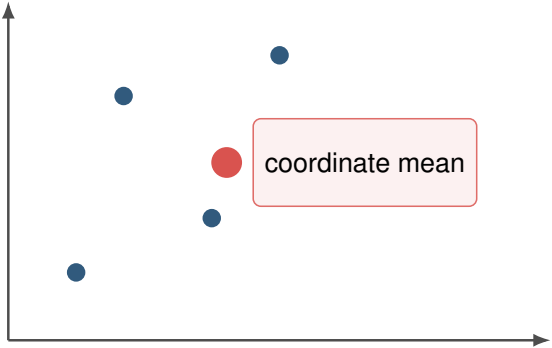
Read it

Voronoi cells partition the feature space.

Boundary

Ties need a deterministic convention for reproducibility.

Update computes the coordinate mean



$$\mu_k \leftarrow \frac{1}{|C_k|} \sum_{i:z_i=k} x_i$$

Read it

The arithmetic mean minimizes squared Euclidean error.

Boundary

For non-Euclidean distances, the mean may no longer be optimal.

Alternating steps never increase the objective

Start from evidence

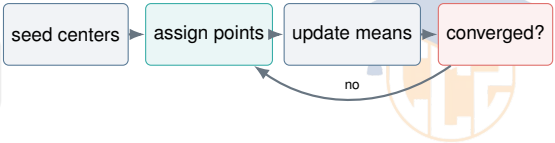
The objective falls until labels stop changing or tolerance is met.

Model statement

Assignment and update each optimize one block while holding the other fixed.

Failure mode

Monotone descent does not guarantee the global minimum.



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Read one complete K-means run

What changes

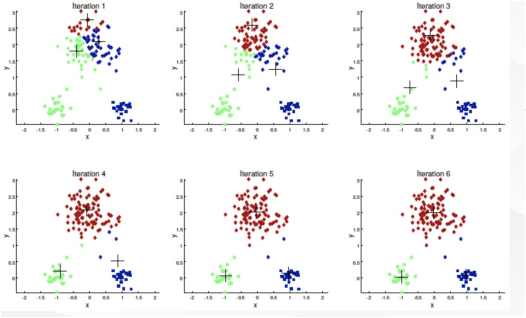
Track centers, assignments, and objective after every iteration.

What can break

A final still image hides initialization sensitivity.

Try this

A three-cluster example stabilizes after several visible moves.



A stopping rule needs a tolerance

Lens	Evidence
Principle	Stop when center movement or objective improvement falls below a threshold.
Case	Also impose a maximum iteration count.
Risk	Exact label equality can be unnecessarily strict with floating-point noise.

The objective curve should drop then flatten

Core idea

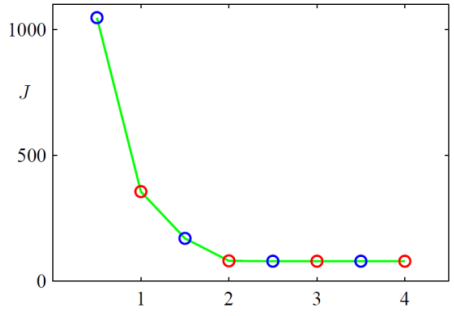
Plot within-cluster sum of squares across iterations.

Concrete case

A sudden increase signals an implementation error.

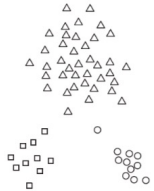
Watch for

A flat curve does not certify a good local optimum.

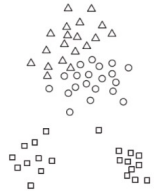


Initialization changes the local optimum

- Different seeds can lead to different final centroids and partitions.
- Run several independent starts and retain the lowest objective.
- Reporting only one favorable seed hides instability.

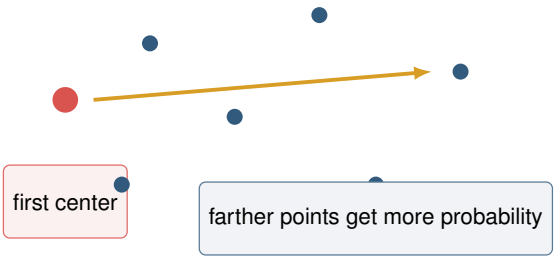


(a) Optimal clustering.



(b) Suboptimal clustering.

K-means++ spreads initial centers



Start from evidence

Far uncovered regions are more likely to receive a center.

Model statement

Choose the next seed with probability proportional to squared distance from existing seeds.

Failure mode

The method improves starts but does not remove local minima.

Empty clusters need an explicit repair

What changes

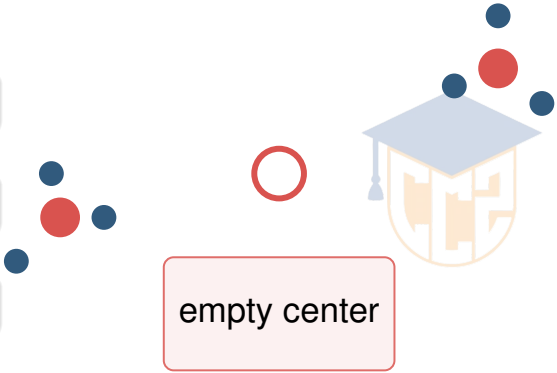
Reinitialize an empty center to a far or high-error point.

What can break

Silently dropping the cluster changes the requested K.

Try this

An unlucky iteration may assign no samples to one center.



Choose K with several clues

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Lens	Evidence
Principle	Combine elbow behavior, silhouette profiles, stability, interpretability, and operational needs.
Case	A warehouse may support only four actionable segments.
Risk	The mathematically best index may imply an unusable number of clusters.



The elbow plot shows diminishing returns

$$W_K = \sum_k \sum_{i \in C_k} \|x_i - \mu_k\|^2$$

Read it

Look for a bend where additional clusters buy little reduction.

Boundary

Many datasets have no clear elbow.

Silhouette can compare candidate K

- Compute sample-level scores for each candidate partition.
- Check both the mean and the distribution by cluster.
- Silhouette tends to prefer well-separated convex groups.

Repeated starts expose solution stability

Start from evidence

A wide spread of objectives indicates a rugged search landscape.

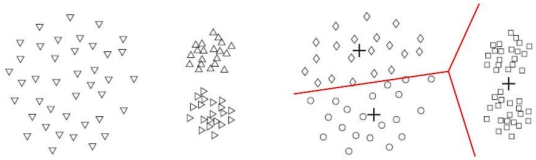
Model statement

Store objective, ARI across runs, and centroid variation.

Failure mode

Selecting one run erases evidence about uncertainty.

K-means prefers similar spherical spread



(a) Original points.

(b) Three K-means clusters.

What changes

Squared Euclidean Voronoi cells work best for compact comparable groups.

What can break

Standardization does not fix unequal population size.

Try this

Clusters of very different size can be cut incorrectly.

Unequal density can mislead centroids

Core idea

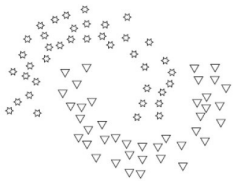
A diffuse cluster may donate points to a nearby dense cluster.

Concrete case

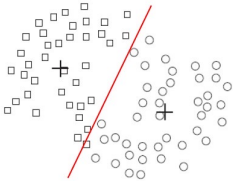
Inspect local density and residual distance by cluster.

Watch for

Increasing K may split density rather than recover meaning.



(a) Original points.



(b) Two K-means clusters.

Nonconvex shapes violate Voronoi geometry

Core idea

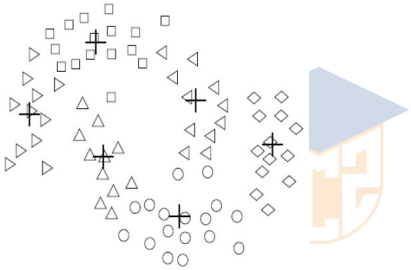
Nearest-centroid regions are convex cells.

Concrete case

A ring or crescent is sliced even when visually coherent.

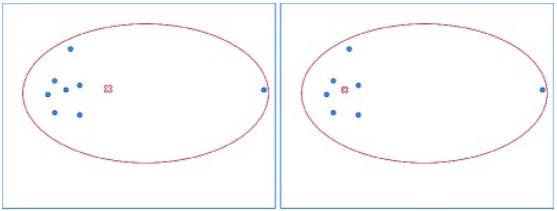
Watch for

More restarts cannot fix a structural mismatch.



(c) Non-spherical shapes.

Outliers pull the arithmetic mean



(a) Mean (b) Medoid

- A distant observation can shift a centroid substantially.
- Compare the mean with a medoid chosen from observed cases.
- Deleting outliers without provenance can remove rare valid segments.

K-medoids uses observed representatives

Start from evidence

It can use arbitrary pairwise distances and is more robust to extremes.

Model statement

Choose a member point minimizing within-cluster dissimilarity.

Failure mode

Optimization is usually more expensive than K-means.

Mini-batch K-means scales to large data

What changes

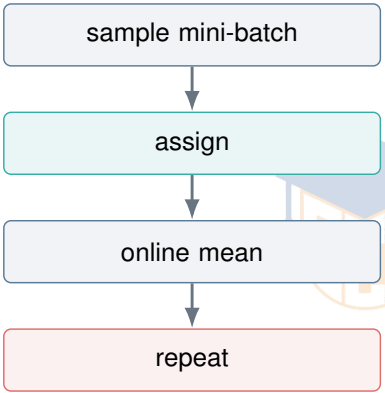
Update centers from small random batches rather than all points each iteration.

What can break

Approximate updates add stochastic variability.

Try this

Useful for millions of moderate-dimensional observations.



Faster updates trade a small amount of objective precision for scale.

Fit K-means with reproducible settings

```
from sklearn.datasets import make_blobs
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
X, _ = make_blobs(n_samples=400, centers=4, random_state=7)
model = make_pipeline(StandardScaler(), KMeans(n_clusters=4, n_init=20, random_state=7))
labels = model.fit_predict(X)
print(model.named_steps['kmeans'].inertia_)
```

Expected observation

Repeat with several K values under the same preprocessing.

Check

Do not refit a different scaler for every plot without recording it.

Profile K-means clusters in original units

Core idea

Inverse-transform centroids and summarize robust distributions of members.

Concrete case

A standardized centroid of 1.2 becomes meaningful only after returning to original units.

Watch for

Centroid averages can hide multimodal members.

A Gaussian mixture is a density model

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k)$$

Read it

Each component contributes a weighted Gaussian density.

Boundary

Components are modeling devices, not guaranteed real populations.

Mixing weights form a probability vector

$$\pi_k \geq 0, \qquad \sum_k \pi_k = 1$$

Read it

The weights estimate component prevalence in the modeled population.

Boundary

Sampling bias changes the apparent weights.

Responsibilities give soft memberships

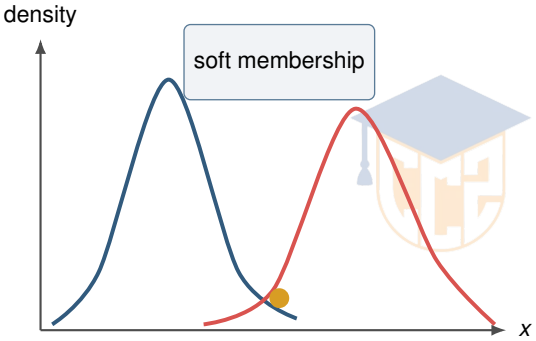
$$r_{ik} = \frac{\pi_k \mathcal{N}(x_i \mid \mu_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(x_i \mid \mu_j, \Sigma_j)}$$

Read it

A point in an overlap region can retain substantial membership in two components.

Boundary

Hardening by argmax discards uncertainty.



Covariance controls component shape



Core idea

Spherical, diagonal, tied, and full covariance models express different geometry.

Concrete case

Full covariance allows rotated ellipses.

Watch for

Flexible covariance needs more data and can become singular.

The E step computes responsibilities

- Hold parameters fixed and evaluate the soft assignment of every point.
- Overlapping components share ambiguous observations.
- Underflow requires stable log-density calculations.

The M step updates weighted parameters

$$N_k = \sum_i r_{ik}, \quad \mu_k = \frac{1}{N_k} \sum_i r_{ik} x_i$$

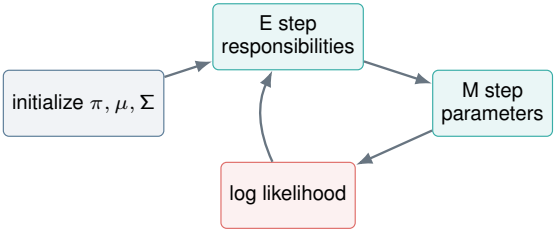
Read it

Covariances and mixture weights use the same responsibility weights.

Boundary

A component with tiny effective count can collapse.

EM alternates two tractable problems



What changes

E and M steps monotonically improve or preserve observed-data likelihood.

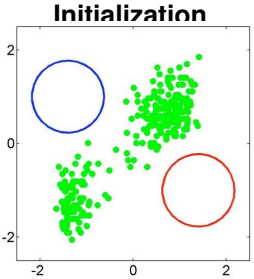
What can break

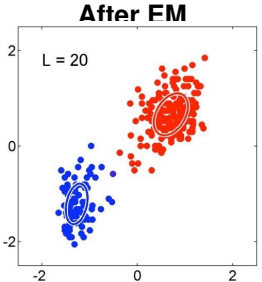
EM converges to a local optimum or saddle, not necessarily the global best.

Try this

Stop when likelihood improvement is small.

Watch a mixture move from seeds to ellipses





Core idea. Initial circles and means evolve into oriented density components.

Concrete case. Color gradients show soft responsibilities during fitting.

Watch for. A final argmax plot hides uncertainty at overlaps.

Log likelihood evaluates fit, not usefulness

$$\ell(\theta) = \sum_i \log \sum_k \pi_k \mathcal{N}(x_i \mid \mu_k, \Sigma_k)$$

Read it

More components can increase training likelihood.

Boundary

Unpenalized likelihood cannot choose K fairly.

AIC and BIC penalize mixture complexity

$$\mathrm{AIC} = 2p - 2\ell, \qquad \mathrm{BIC} = p \log N - 2\ell$$

Read it

Lower values balance fit with parameter count under different penalties.

Boundary

Singular fits and poor optimization can invalidate comparison.

Multiple initializations remain necessary

Start from evidence

K-means initialization often provides a useful start.

Model statement

Run EM from several seeds and retain high-likelihood stable solutions.

Failure mode

The highest likelihood can still contain a nearly singular component.

Regularize covariance estimates

$$\Sigma_k \leftarrow \Sigma_k + \lambda I$$

Read it

A small diagonal term prevents zero-variance collapse.

Boundary

Too much regularization erases genuine anisotropy.

Agglomerative clustering starts from singletons

Core idea

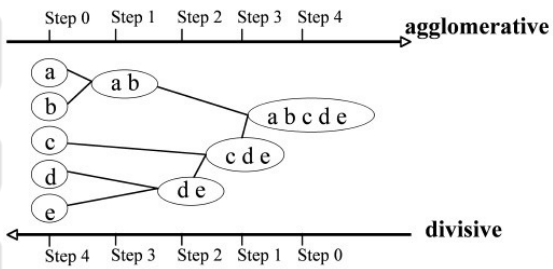
Treat every object as its own cluster and repeatedly merge the closest pair.

Concrete case

The process yields a nested sequence of partitions.

Watch for

Early merges cannot be undone later.



Divisive clustering starts from one group

- Recursively split the full dataset into smaller groups.
- Useful when top-level separation is easier to define.
- Exact optimal splitting is generally expensive.

Linkage defines distance between clusters

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single
nearest pair

complete
farthest pair

average
all pairs

chaining

compact groups

middle ground

Start from evidence

The same point distance can yield different dendrograms.

Model statement

Single, complete, average, and Ward linkage encode different shapes.

Failure mode

Linkage is a modeling choice, not a plotting option.

Single linkage uses the nearest pair

$$d(A, B) = \min_{i \in A, j \in B} d(i, j)$$

Read it

It can recover elongated connected groups.

Boundary

A few bridge points create chaining.

Complete linkage uses the farthest pair

$$d(A, B) = \max_{i \in A, j \in B} d(i, j)$$

Read it

It favors compact clusters with controlled diameter.

Boundary

It can split elongated natural groups.

Average linkage uses all cross pairs

$$d(A, B) = \frac{1}{|A||B|} \sum_{i \in A, j \in B} d(i, j)$$

Read it

It balances local bridges and worst-case diameter.

Boundary

Computation still relies on the chosen base distance.

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Ward linkage minimizes variance increase

- Merge the pair causing the smallest rise in within-cluster sum of squares.
- It often produces compact balanced groups under Euclidean geometry.
- Ward is not compatible with arbitrary dissimilarities.

A dendrogram records merge heights

Start from evidence

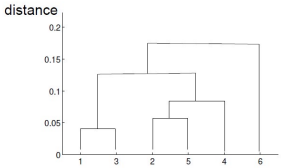
Read bottom-up to see when groups join.

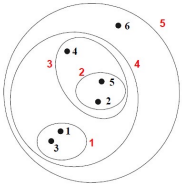
Model statement

Leaves are objects and branch heights are linkage dissimilarities.

Failure mode

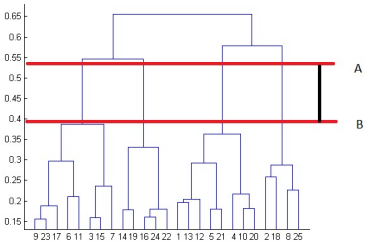
Vertical spacing is meaningful; horizontal leaf order often is not.





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Cut height selects a partition



What changes

A horizontal cut intersects one branch for each chosen cluster.

What can break

A large visual gap is a clue, not proof of the correct cut.

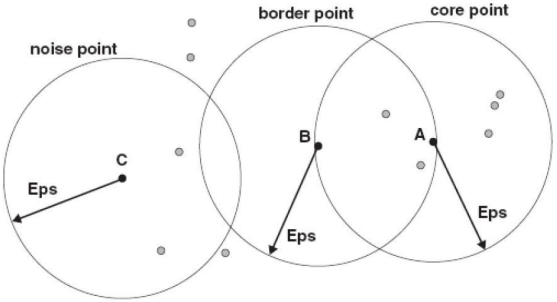
Try this

Move the cut to inspect coarse and fine resolutions.

Hierarchical clustering scales quadratically or worse

Lens	Evidence
Principle	Pairwise storage and repeated updates limit very large datasets.
Case	Use sampling, connectivity constraints, or approximate neighbors when needed.
Risk	A readable dendrogram of thousands of leaves is not informative.

DBSCAN uses a neighborhood radius and count



- Parameters are ϵ and MinPts.
- A point is core when its radius neighborhood is sufficiently populated.
- The same numeric radius changes meaning after scaling.

Core points can expand a cluster

$$|N_\epsilon(x_i)| \geq \mathrm{MinPts}$$

Read it

Density-reachable core neighborhoods are connected into one cluster.

Boundary

Counting the point itself must follow the implementation convention.

Border points join but do not expand

What changes

A border point lies near a core point but lacks enough neighbors itself.

What can break

A border shared by two clusters can depend on traversal conventions.

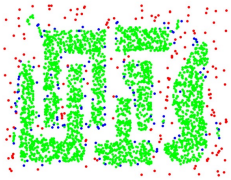
Try this

It receives the cluster label of reachable density structure.

Noise points remain unassigned



Original Points



Point types: core, border and noise

Eps = 10, MinPts = 4

Core idea

A point that is neither core nor reachable border receives the noise label.

Concrete case

Rare anomalies can be separated from dense groups.

Watch for

Noise does not necessarily mean error or irrelevance.

Cluster expansion follows density reachability

Core idea

Start from an unvisited core, add its neighbors, and recursively expand through new cores.

Concrete case

Continue until no new density-reachable point remains.

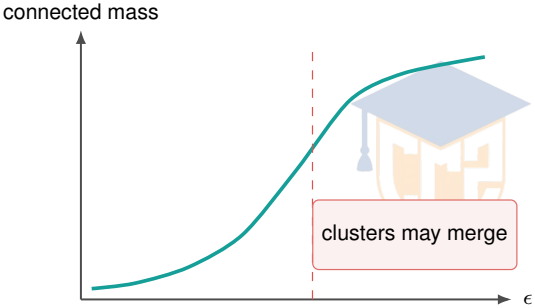
Watch for

Starting from a border point should not launch independent expansion.



Epsilon controls neighborhood connectivity

- Small ϵ fragments groups and creates noise; large ϵ merges groups.
- Trace cluster count and noise fraction across a range.
- A single global radius struggles with variable densities.



MinPts controls density evidence

Start from evidence

A rough starting point often relates to feature dimension.

Model statement

Larger MinPts demands a more strongly supported local neighborhood.

Failure mode

Rules of thumb are not substitutes for validation.

A k-distance curve suggests epsilon

What changes

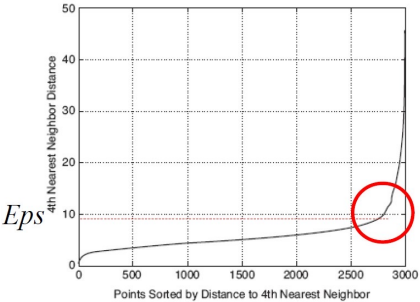
Sort each point's distance to its kth nearest neighbor and inspect the bend.

What can break

The elbow may be unclear under multiple densities.

Try this

Set k consistently with the MinPts convention.



Variable densities are a central limitation

Lens	Evidence
Principle	One radius may split a sparse cluster while merging dense neighbors.
Case	Compare DBSCAN with OPTICS or HDBSCAN-like approaches when allowed.
Risk	Changing algorithms also changes the clustering definition.



Fit DBSCAN and inspect all point roles

```
from sklearn.datasets import make_moons
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import DBSCAN
X, _ = make_moons(n_samples=400, noise=0.06, random_state=7)
model = make_pipeline(StandardScaler(), DBSCAN(eps=0.35, min_samples=6))
labels = model.fit_predict(X)
print('clusters', len(set(labels)) - (1 if -1 in labels else 0))
print('noise rate', np.mean(labels == -1))
```

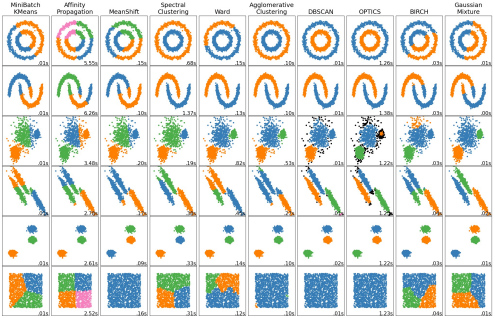
Expected observation

Sweep radius values and retain cluster and noise summaries.

Check

Silhouette is undefined or misleading when almost everything is noise.

Algorithm comparison reveals structural bias

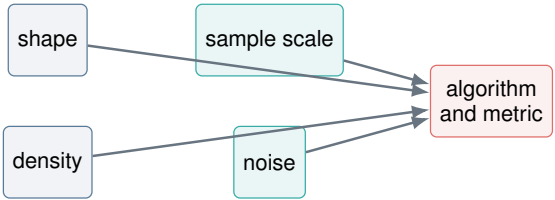


Start from evidence. Read each row as a data geometry and each column as an algorithm assumption.

Model statement. K-means, spectral, linkage, DBSCAN, OPTICS, and mixtures behave differently across shapes.

Failure mode. A colorful benchmark is illustrative, not a universal ranking.

Choose from shape, density, scale, and purpose



What changes

Center-based methods fit compact prototypes; mixtures fit ellipses; linkage gives hierarchy; DBSCAN follows density connectivity.

What can break

Trying every algorithm and reporting one winner inflates selection bias.

Try this

Start with two or three assumption-matched candidates.

A leak-safe clustering workflow

Core idea

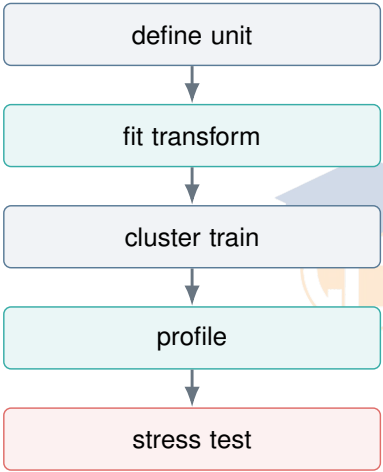
Fit transformations on a reference training sample, freeze them, and test assignment or stability on new data.

Concrete case

Profile clusters in original units and monitor membership drift.

Watch for

Refitting the entire pipeline every month changes the meaning of cluster IDs.



Unsupervised does not mean unevaluated.

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Open-book case: customer segments

Core idea

A four-cluster K-means solution has a high silhouette but changes after replacing Euclidean with cosine distance.

Concrete case

Explain the ambiguity and propose a metric-grounded stability study.

Watch for

A strong answer links the distance to the business meaning of magnitude versus profile.

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Clustering self-check

1. State why clustering has no unique answer without assumptions.
2. Choose a distance for one numeric, text, or mixed dataset.
3. Combine internal, external, and stability evidence.
4. Diagnose one K-means failure.
5. Contrast GMM, hierarchy, and DBSCAN.

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A cluster is a group under a stated similarity rule

- Changing features or distance can change the groups.
- A cluster number is an identifier, not an ordered score.
- Good geometric separation does not automatically imply useful categories.
- Describe what each group means and where the result is stable.